COMPSYS 305 Mini-Project – Interim Design Report

1. Game Strategy and Design Specifications

The project implements a Flappy Bird-style side-scrolling game on the Terasic DE0-CV FPGA board (Cyclone V), displayed on a 640×480 VGA screen. The player controls a bird that falls under gravity and must be flapped upward to navigate through gaps in oncoming pipe obstacles. The game provides two modes: Training (fixed slowest speed, infinite lives) and Play (10 levels of increasing difficulty (speed up the pipes). The bird dies if it touches the ground, ceiling, any pipe, or runs out of lives.

1.1 Control Scheme

- **Mouse:** Mouse left-click: flap bird upward (applies upward impulse of 4 px/frame, resetting fall acceleration)
- **KEY[0]:** KEY[0] (push-button): active-low reset — returns bird to vertical centre (row 240) and resets timer.
- **KEY[1]:** KEY[1] (push-button): pause/resume toggle — freezes bird physics, timer, and starfield.
- **SW[1]:** SW[1]: toggle title text overlay on/off (demonstrates DIP switch functionality).
- **SW[3–8]:** SW[3]–SW[8]: bird colour selector (red / orange / yellow / green / blue / purple; default white if none active).

1.2 Difficulty and Levels

Level progression will be based on the score the player has (shown by an on screen score counter). At each new level, pipe scroll speed increases by 4 px/frame and minimum pipe gap decreases. Ten levels are planned: you will increment a level each 10+ points to the score . In Training mode the level is fixed at level 1 and has infinite lives.

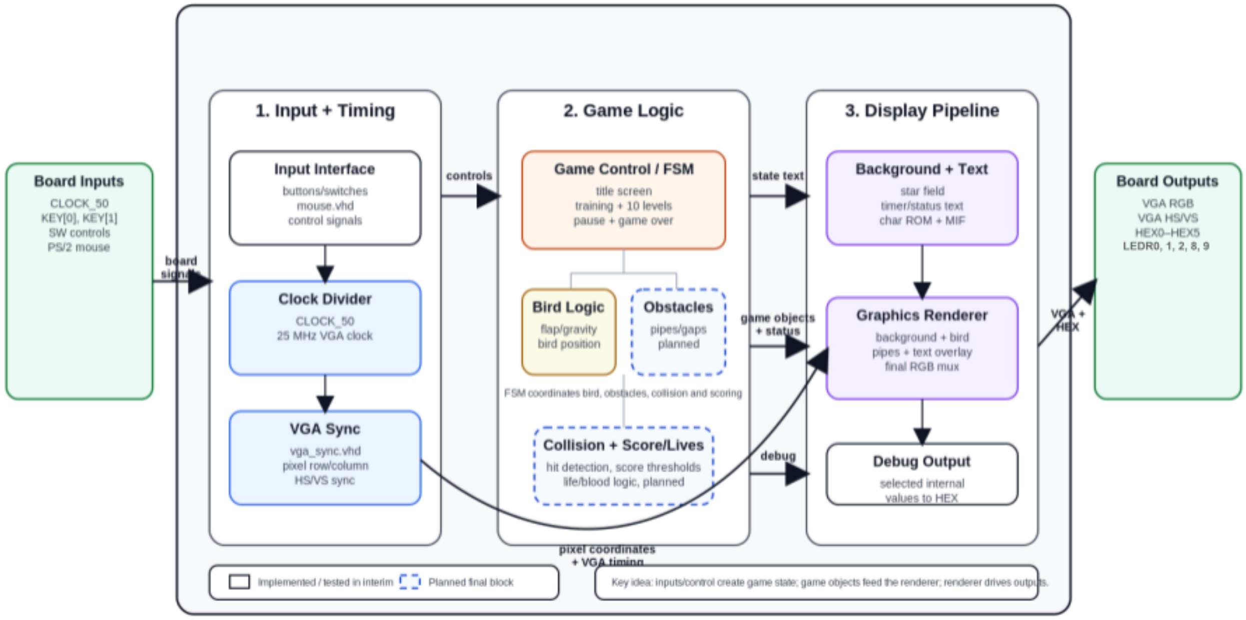
1.3 Scoring and Lives

Score increments each time the bird passes through a pipe gap. Lives start at 3; contact with an obstacle deducts one life. The current score and lives will be displayed on the seven-segment displays and on-screen as text. (Timer is already rendered on the VGA display using the char_rom module at 2× scale.)

1.4 Current Implementation Status

- VGA sync and pixel generation (VGA_SYNC.vhd) — complete.
- PS/2 mouse driver (MOUSE.vhd) — complete; left/right button states, cursor row and column available. Mouse cursor column/row currently shown on HEX2–HEX5 seven-segment displays.
- Bird physics: gravity with capped fall speed (6 px/frame), left-click flap, ceiling/ground clamping — complete.
- Bird colour selection via SW[3:8] — complete.
- Pause/resume via KEY[1] (falling-edge detect) — complete; LEDR[2] indicates pause state.
- Reset via KEY[0] — complete; bird returns to row 240.
- Scrolling 200-star pseudo-random starfield (LFSR, star_field.vhd) — complete.
- On-screen up-counting timer 0:00–9:59 at 2× scale using char_rom (screen_timer.vhd) — complete.
- Title text 'SUS BIRD' at mixed scale (2× for 'SUS', 1× for 'BIRD') toggled by SW[1] and 'PAUSED' overlay via char_rom — complete.
- Seven-segment displays: bird Y position on HEX0–HEX1; mouse column on HEX2–HEX3; mouse row on HEX4–HEX5 — complete.
- LEDR[0,1,2,8] = left mouse click, right mouse click, paused state, bird falling — complete
- Pipes, Sprites/design, collision detection, score counter, lives display, game-over screen, Title screen, *countdown state, *audio — planned for final submission.

2. System Block Diagram



3. High-Level Game State Machine